MetaBeam ROS Driver

1. Environment and Dependencies

System environment requirement: Linux + ROS
Recommanded: Ubuntu 16.04 - with ROS kinetic desktop-full installed or
Ubuntu 18.04 - with ROS melodic desktop-full installed or
Ubuntu 20.04 - with ROS noetic desktop-full installed.

Check resources on http://ros.org for installation guide.

2. Installation

```
sudo apt-get update
sudo apt-get install ros-<your-ros-distro>-velodyne
```

2.1. Install dependencies

2.2. Create the workspace for Driver

```
cd~
mkdir -p catkin_ws/src
```

2.3. Driver source code installation

```
cd ~/catkin_ws/src
git clone https://github.com/ros-drivers/velodyne.git
```

2.4. Build driver

```
cd ~/catkin_ws
catkin_make
```

3. Configure PC IP Address

Before using MetaBeam, you need to configure its parameters, including the IP address, port, and model. MetaBeam connects to the computer via Ethernet by default, so you need to ensure that both the computer and the LIDAR are on the same local network.

You can set the computer's IP address by modifying the following command:

```
sudo ifconfig    <your-network-port-name> 192.168.1.200
```

4. View the Real Time Data

4.1. Convert xml file to yaml file

Move the LK-32.xml file to the velodyne_pointcloud/params directory.

```
mv /path/to/LK-32.xml /path/to/velodyne_pointcloud/params/
```

4.2. Modify the launch file

Change line 7 of the VLP32_points.launch file to the following content.

```
<arg name="calibration" default="$( find
velodyne_pointcloud)/params/LK-32-final.yaml"/>
```

4.3. Ros launch

```
roslaunch velodyne_pointcloud VLP-32C_points.launch
```

4.4. Ros rviz

```
rosrun rviz rviz -f velodyne
```